Machine Learning Algorithm Comparison Report

End-to-End Evaluation of Clustering, Classification, and Boosting Paradigms

1. Executive Summary & Objectives

This project implements a comprehensive comparative study across three distinct machine learning tasks using three real-world datasets: Mall_Customers.csv (Customer Segmentation via Clustering), Social_Network_Ads.csv (Purchase Prediction via Supervised Classification), and insurance_pre.csv (Lifestyle & Risk Modeling via Sequential Ensemble Boosting). All algorithms were executed with leakage-free preprocessing pipelines and evaluated using standardized quantitative metrics.

2. Best-Performing Algorithms Summary

Task / Dataset	Best Algorithm	Primary Metric	Score Achieved
Customer Segmentation (Clustering)	K-Means (k=5)	Silhouette Score & Full-Population Coverage	Silhouette: 0.5547 (Calinski-Harabasz: 248.65
Purchase Prediction (Classification)	K-Nearest Neighbors	F1-score (Balance of Precision & Recall)	F1: 0.8767 (Accuracy: 91.0%, ROC-AUC: 0.9
Lifestyle Risk Prediction (Boosting)	Gradient Boosting	F1-score & ROC-AUC	F1: 0.9369 (Accuracy: 97.39%, ROC-AUC: 0.9

3. Unsupervised Clustering (Mall Customers)

K-Means, Agglomerative Clustering (Ward linkage), and DBSCAN were evaluated on standardized Annual Income and Spending Score. K-Means (k=5) achieved the highest Silhouette Score of 0.5547 and lowest Davies-Bouldin Index of 0.5726, cleanly identifying 5 customer archetypes.

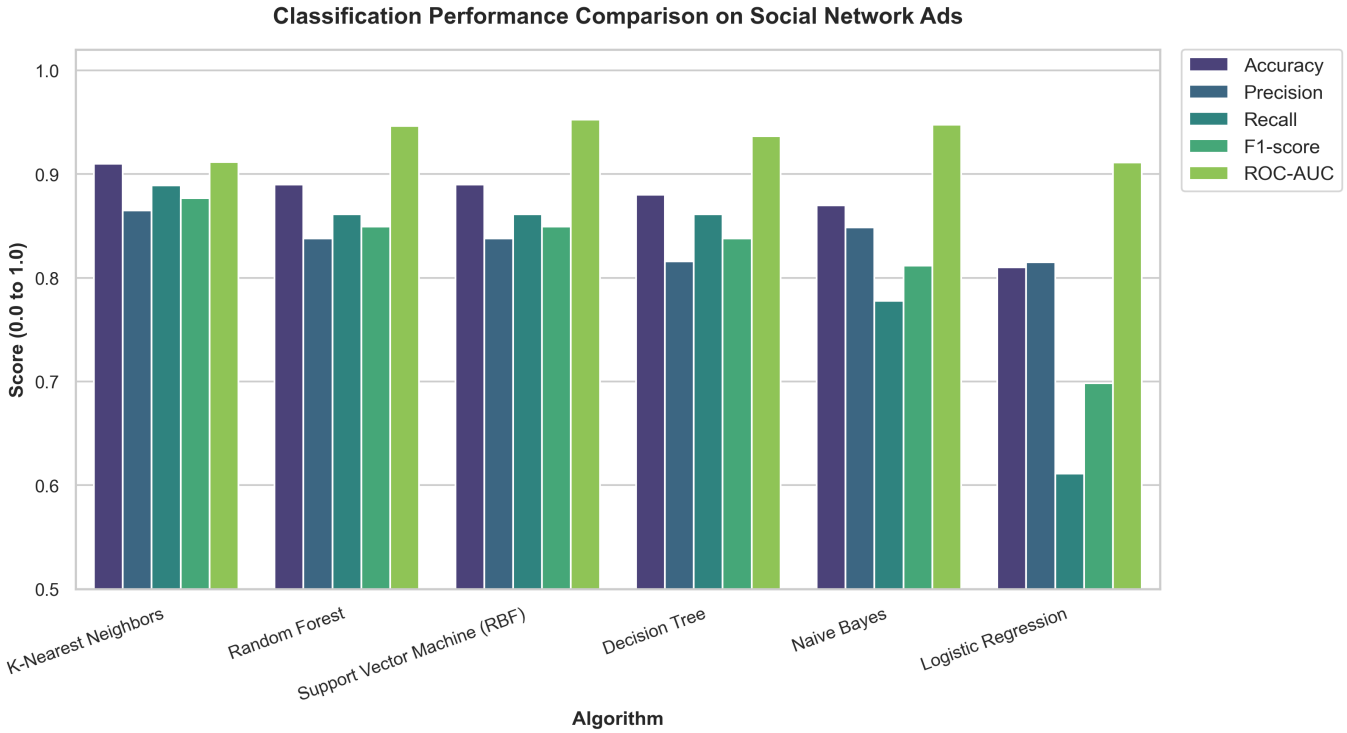
Algorithm	Clusters	Noise	Silhouette	Davies-Bouldin	Calinski-Harabasz
K-Means	5	0	0.5547	0.5722	248.65
Agglomerative (Hierarchical)	5	0	0.5538	0.5779	244.41
DBSCAN	6	23	0.5577	0.5106	245.36



4. Supervised Classification (Social Network Ads)

Six classifiers were trained using stratified 5-fold cross-validation and evaluated on an unseen test set (25% holdout). K-Nearest Neighbors (k=5) and SVM (RBF) achieved superior decision boundary resolution for non-linear age/salary thresholds compared to linear Logistic Regression.

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC	5-Fold CV Acc
K-Nearest Neighbors	0.9100	0.8649	0.8889	0.8767	0.9117	0.8967
Random Forest	0.8900	0.8378	0.8611	0.8493	0.9462	0.9067
Support Vector Machine (RBF)	0.8900	0.8378	0.8611	0.8493	0.9527	0.9100
Decision Tree	0.8800	0.8158	0.8611	0.8378	0.9364	0.9133
Naive Bayes	0.8700	0.8485	0.7778	0.8116	0.9475	0.8833
Logistic Regression	0.8100	0.8148	0.6111	0.6984	0.9110	0.8400



5. Sequential Ensemble Boosting (Insurance Dataset)

AdaBoost, Gradient Boosting, and XGBoost were evaluated for predicting policyholder smoker status. Gradient Boosting achieved 97.39% accuracy, an F1-score of 0.9369, and an ROC-AUC of 0.9952. Feature importance analysis confirmed that medical charges billed is the dominant predictor.

Algorithm	Target	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Gradient Boosting	smoker	0.9739	0.9286	0.9455	0.9369	0.9952
AdaBoost	smoker	0.9701	0.9123	0.9455	0.9286	0.9930
XGBoost	smoker	0.9664	0.9107	0.9273	0.9189	0.9927

